

Given: $S_1 = (3, 2)$, $S_2 = (-1, 1)$, $S_3 = (4, 0)$

To show linear independence, find a nontrivial solution to $aS_1 + bS_2 + cS_3 = (0, 0)$

This gives us the system $a(3, 2) + b(-1, 1) + c(4, 0) = (0, 0)$

$$\rightarrow \begin{cases} 3a - b + 4c = 0 \\ 2a + b + 0c = 0 \end{cases} \quad \text{matrix: } \left[\begin{array}{ccc|c} 3 & -1 & 4 & 0 \\ 2 & 1 & 0 & 0 \end{array} \right]$$

$$1.) \frac{1}{3}R_1 \rightarrow R_1 \quad \left[\begin{array}{ccc|c} 1 & -\frac{1}{3} & \frac{4}{3} & 0 \\ 2 & 1 & 0 & 0 \end{array} \right]$$

$$2.) -2R_1 + R_2 \rightarrow R_2 \quad \left[\begin{array}{ccc|c} 1 & -\frac{1}{3} & \frac{4}{3} & 0 \\ 0 & \frac{5}{3} & -\frac{8}{3} & 0 \end{array} \right]$$

$$3.) \frac{3}{5}R_2 \rightarrow R_2 \quad \left[\begin{array}{ccc|c} 1 & -\frac{1}{3} & \frac{4}{3} & 0 \\ 0 & 1 & -\frac{8}{5} & 0 \end{array} \right]$$

$$4.) \frac{1}{3}R_2 + R_1 \rightarrow R_1 \quad \left[\begin{array}{ccc|c} 1 & 0 & \frac{4}{5} & 0 \\ 0 & 1 & -\frac{8}{5} & 0 \end{array} \right]$$

So, $a + \frac{4}{5}c = 0 \rightarrow a = -\frac{4}{5}c$, $b - \frac{8}{5}c = 0 \rightarrow b = \frac{8}{5}c$

Let $c = 5$, then $a = -4$, $b = 8$, $c = 5$

Therefore, $(0, 0) = -4S_1 + 8S_2 + 5S_3$

Express S_1 as a linear combination of S_2 and S_3 :

From $-4S_1 + 8S_2 + 5S_3 = 0$:

$$-4S_1 = -8S_2 - 5S_3 \rightarrow S_1 = 2S_2 + \frac{5}{4}S_3$$